# Lecture 17 — Linear Programming (Part 1)

## COMPSCI 308 — Design and Analysis of Algorithms

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Jan-Mar 2026

# Summary

ITA 29 Linear Programming

ITA 29.1 Standard and Slack Forms

# Assignments[1]

Practice makes perfect!

Introduction to Algorithms (ITA) 📖 Required Readings:

- Section 29.1.

✏️ Required Exercises:

- Exercises 29.1 — 1–8.

[1] 💣 Assignments will not be collected; however, quiz problems will be selected from them. (This includes both Readings and Exercises.)

# ITA 29 Linear Programming

# An Election Problem

Imagine you are a campaign manager for an upcoming election.

The numbers of voters are — 100k 🏙️, 200k 🏘️, 50k 🏞️.

At least 50k, 100k, and 25k 🗳️ from each group are needed.

Determine the least amount of ad money needed.

| Policy | 🏙️ | 🏘️ | 🏞️ |
|---|---|---|---|
| Prep 🧑‍💼 | -2k 👎 | 5k 👍 | 3k 👍 |
| Sharks 🦈🔦 | 8k 👍 | 2k 👍 | -5k 👎 |
| Flying cars 🚗🛩️ | 0k 😐 | 0k 😐 | 10k 👍 |
| Dolphin vote 🐬🗳️ | 10k 👍 | 0k 😐 | -2k 👎 |

**Table 1:** Effects on voters per $1000 spent on ads

# A Mathematical Model

Let $x_1, x_2, \ldots, x_4$ be the amount of 💵 you are going to spend on ads for 👩‍🦰, 🐊, 🚗 and 🐬.

Then you want to minimize the linear objective function

$$x_1 + x_2 + x_3 + x_4$$

while satisfying the following linear constraints

spend $x_1$ k \$ on zombie prop. will cost you 2k city votes

Total city vote

$$\Rightarrow -2x_1 + 8x_2 + 0x_3 + 10x_4 \geq 50$$

suburb.

$$\Rightarrow 5x_1 + 2x_2 + 0x_3 + 0x_4 \geq 100$$

country side.

$$\Rightarrow 3x_1 - 5x_2 + 10x_3 - 2x_4 \geq 25$$

This type of problem is called an Linear Programming (LP) problem.

# A Second Example

Suppose that to get at least 8 units of vitamin A and 10 units of vitamin C, we eat —

| Food | Vitamin A | Vitamin C | Price (per kg) |
|---|---|---|---|
| 🥕 | 2 | 1 | \$5 👍 |
| 🍊 | 1 | 2 | \$7 👍 |

$\rightarrow x_1$

$\rightarrow x_2.$

**Table 2:** Nutritional values of 🥕 and 🍊

We want to minimize the cost of the food combination. $5 \cdot x_1 + 7\, x_2$

How to formulate this as an LP problem?

vitamin A : $2 \cdot x_1 + 1 \cdot x_2 \geq 8$

vitamin C $1 \cdot x_1 + 2\, x_2 \geq 10.$

$x_1, x_2 \geq 0.$

# 💬 Wooclap — Balancing Studies and Gaming

You're a student with 20 hours per week for reading 📚 and playing 🕹️.

$x_1$ $x_2$

Reading 📚 is 10 times more enjoyable than playing 🕹️.

You want to maximize total enjoyment given the constraints —

- Minimum ~~study~~ reading time 5 hours.
- Minimum gaming time 10 hours.

objective function.

Maximize : $10 \cdot x_1 + x_2$

constraints: linear inequality

$x_1 \geq 5$

$x_2 \geq 10$

$x_1 + x_2 = 20$

linear equality.

Optimal (the one maximizes objective is

$x_1 = 10$

$x_2 = 10$

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You're a student with 20 hours per week for reading 📚 and playing 🕹️.

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You want to maximize total enjoyment given the constraints —

- Minimum study time 5 hours.
- Minimum gaming time 10 hours.

Let us try to formulate this as a LP problem —

$$
\begin{aligned}
\text{maximize} \quad & \underline{\qquad\qquad} \\
\text{subject to} \quad & \underline{\qquad\qquad} \geq 5 \\
& \underline{\qquad\qquad} \geq 10 \\
& \underline{\qquad\qquad} \leq 20
\end{aligned}
$$

where $x_1$ is the study time and $x_2$ is the game time.

# Linear Programming Problem

Linear Programming (LP) optimizes a linear objective function subject to linear constraints.

Common applications include —

- 💰 Resource allocation in businesses for cost minimization.
- 📈 Optimizing a portfolio of investments to maximize total return.
- 🍳 Diet optimization for nutritional balance at minimal cost.
- 🧪 Optimizing product mix in manufacturing to maximize profit.

# Linear Function

A linear function $f$ of variables $x_1, x_2, \dots, x_n$ with coefficients $a_1, a_2, \dots, a_n$ is given by —

$$f(x_1, x_2, \dots, x_n) = \sum_{j=1}^{n} a_j x_j.$$

Linear combination of

$x_1, x_2, \dots x_n.$

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$$f(x_1, x_2, \dots, x_n) = \sum_{j=1}^{n} a_j x_j.$$

A linear equality is of the form $f(x_1, x_2, \dots, x_n) = b$, where $b$ is a real number.

Linear inequalities take the form —

$$f(x_1, x_2, \dots, x_n) \leq b \qquad \text{or} \qquad f(x_1, x_2, \dots, x_n) \geq b.$$

These are collectively called linear constraints or simply constraints.

# 📌 Example of Linear Program

Consider the following LP problem with two variables —

$$
\begin{aligned}
\text{maximize} \quad & x_1 + x_2 \\
\text{subject to} \quad & 4x_1 - x_2 \leq 8 \\
& 2x_1 + x_2 \leq 10 \\
& 5x_1 - 2x_2 \geq -2 \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

$x_1 \geq 0$

$x_2 \geq 0$

🤔 How can we solve this?

# The Feasible Area

The intersection of the constraints

$$
\begin{aligned}
4x_1 - x_2 &\leq 8 \\
2x_1 + x_2 &\leq 10 \\
5x_1 - 2x_2 &\geq -2 \\
x_1, x_2 &\geq 0.
\end{aligned}
$$

forms the feasible area.

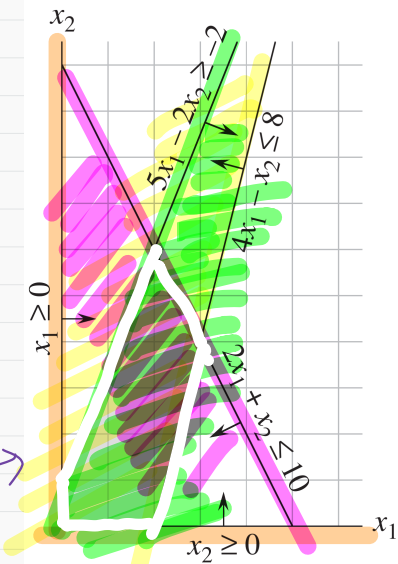


**Figure 1:** The feasible area

# The Objective Value

The value of the objective function

$$x_1 + x_2$$

is called the objective value.

We could compute it at every point in the feasible area to find the optimal value.

🤔 How can we be more efficient?

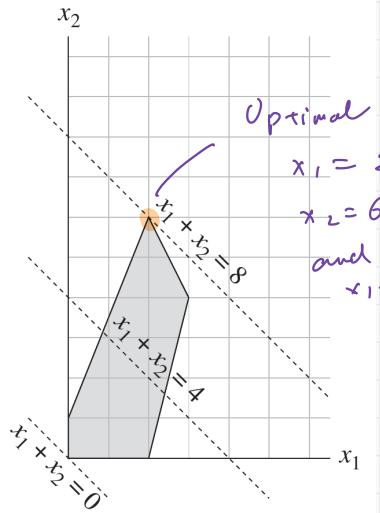


**Figure 2:** The feasible area

# 💬 Wooclap Exercise 29.1-4

infeasible.

🍰 What is the feasible area for the following?

$$\begin{aligned} \text{maximize} \quad & 3x_1 - 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 2 \\ & -2x_1 - 2x_2 \leq -10 \\ & x_1, x_2 \geq 0. \end{aligned}$$

No $x_1, x_2$ satisfies both.

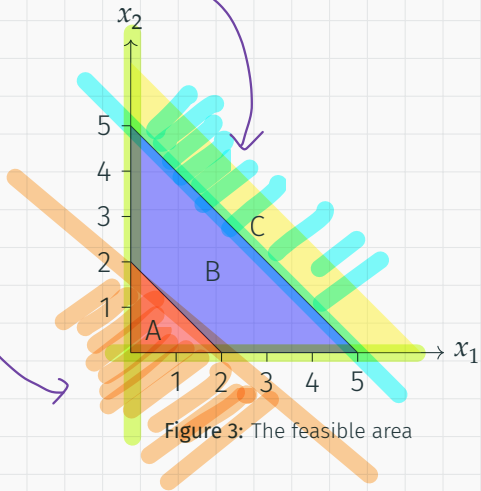


**Figure 3:** The feasible area

# Linear Programming Methods

Three key methods —

- Simplex 🚀 — Efficient in practice but exponential on worst cases.
- Ellipsoid 🐌 — Polynomial-time but slow in practice.
- Interior-point 🎯 — Polynomial-time, often faster for large inputs.

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**📗 Simplex**

A simplex is the generalization of a 📐 or tetrahedron to arbitrary dimensions.

# Integer Linear Programming (ILP)

## What is Integer Linear Programming (ILP)?

- Similar to LP, but with the added constraint that some or all variables are required to be *integers*.

## Key Complexity Differences —

- LP — Typically solvable in polynomial time.
- ILP — *NP-hard*. (Week 7)

# Integer Linear Programming (ILP)

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## Key Complexity Differences —

- LP — Typically solvable in polynomial time.
- ILP — *NP-hard*. (Week 7)

🗺️ Traveling Salesman Problem

🎒 0-1 Knapsack Problem

🕒 Scheduling Selection Problems

## Applications of ILP — ILP Software

- JuMP (Julia)
- OR-Tools
- Gurobi
- CPLEX
- GLPK
- HiGHS

# Feasible Area with Integer Points

Consider the following problem with two variables —

$$\begin{aligned} \text{maximize} \quad & x_1 + x_2 \\ \text{subject to} \quad & 4x_1 - x_2 \leq 8 \\ & 2x_1 + x_2 \leq 10 \\ & 5x_1 - 2x_2 \geq -2 \\ & x_1, x_2 \geq 0. \end{aligned}$$

The feasible area is for the LP is the shaded region, but ILP requires integer points.

This is why ILP is often much harder to solve.

The optimal solution for the ILP is still here.

$\begin{cases} x_1 = 2 \\ x_2 = 6 \end{cases}$

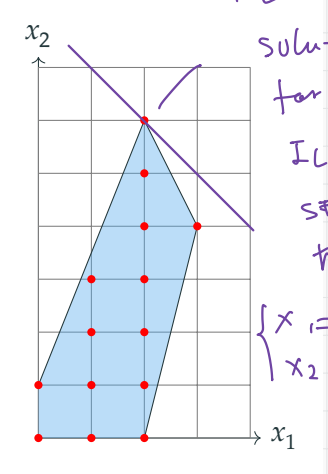


**Figure 4:** What is the optimal objective value for this ILP?

# 🤪 Solving an LP/ILP With computer

Consider the following problem with two variables —

$$
\begin{aligned}
\text{maximize} \quad & x_1 + x_2 \\
\text{subject to} \quad & 4x_1 - x_2 \leq 8 \\
& 2x_1 + x_2 \leq 10 \\
& 5x_1 - 2x_2 \geq -2 \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

To solve this with 💻, one can use JuMP (Julia).

```
using JuMP, HiGHS

model = Model(HiGHS.Optimizer)
@variable(model, x1 >= 0)
@variable(model, x2 >= 0)
@objective(model, Max, x1 + x2)
@constraint(model, 4x1 - x2 <= 8)
@constraint(model, 2x1 + x2 <= 10)
@constraint(model, 5x1 - 2x2 >= -2)

optimize!(model)
```

Channing the solver to `Ipopt` solves the corresponding ILP problem.

# ITA 29.1 Standard and Slack Forms

# Canonical Forms of Linear Programs

We will use two canonical forms of linear programming problems —

- Standard Form — *Maximizing* a linear objective function subject to linear *inequality* constraints.

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# Canonical Forms of Linear Programs

We will use two canonical forms of linear programming problems —

- Standard Form — *Maximizing* a linear objective function subject to linear *inequality* constraints.
- Slack Form — *Maximizing* a linear objective function but with linear *equalities* constraints.

Standard form is widely used for representing linear programs.

Slack form is handy when using the simplex method.

**Slack**

Slack is the extra time after midnight before the homework gets done.

# Standard Form

In standard form, we have the following components:

- A vector $c = (c_1, c_2, \dots, c_n)$ known as cost coefficients.
- A vector $b = (b_1, b_2, \dots, b_m)$ representing the resource limits.
- A matrix $A = (a_{ij})_{1 \le i \le m, 1 \le j \le n}$ representing the constraints.

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The 🎯 is to find $(x_1, x_2, \dots, x_n)$ that

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} x_j \le b_i \quad \text{for } i = 1, 2, \dots, m \\ & x_j \ge 0 \quad \text{for } j = 1, 2, \dots, n. \end{aligned}$$

We will denote an LP problem by $(A, b, c)$.

# Example of Standard Form

What are $c, b, A$ for the following problem?

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\
\text{subject to} \quad & x_1 + x_2 - x_3 \leq 7, \\
& -x_1 - x_2 + x_3 \leq -7, \\
& x_1 - 2x_2 + 2x_3 \leq 4, \\
& x_1, x_2, x_3 \geq 0.
\end{aligned}
$$

$c = (2, -3, 3)$, $b = (7, -7, 4)$.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & -2 & 2 \end{bmatrix}$$

$\Longrightarrow$ Simpler for coding on computer.

# Matrix Form of Standard Form

In other words, we seek a vector $x = (x_1, x_2, \dots, x_n)$

$$\begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}$$

where $c_1 x_1 + c_2 x_2 \cdots c_n x_n$.

- $c^T x = c \cdot x$ is the objective function,
- $Ax \leq b$ and $x \geq 0$ are the constraints,
- $x \geq 0$ are the nonnegativity constraints.

An LP problem does not necessarily have the nonnegativity constraints.

💡 But the *standard form* requires them.

# Linear Programming Solutions

A solution to a linear program is an assignment of values to the variables.

Linear programs can have different types of solutions —

- ✅ Feasible solution — Satisfies all constraints.
- ❌ Infeasible solution — Violates at least one constraint.
- 🏆 Optimal solution — A feasible solution that maximizes (or minimizes) the objective function, achieving the optimal objective value.

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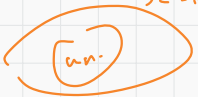
Thus there are three types of LP problems —

- ✅ Feasible problem — Feasible solutions exist.
- ❌ Infeasible problem — No feasible solutions exist.
- ♾ Unbounded problem — Feasible solutions exist, but the objective function is unbounded.

feasible

max $x_1 + x_2$

$x_1 \geq 0, \; x_2 \geq 0.$



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We also say that —

- 🔃 Two *maximization* problems $L$ and $L'$ are equivalent if every solution to $L$ with value $z$ corresponds to a solution to $L'$ with the same value $z$, and vice versa.
- 🔄 A *maximization* problem $L$ and a *minimization* problem $L'$ are equivalent if every solution to $L$ with value $z$ corresponds to a solution to $L'$ with value $-z$, and vice versa.

# Standardization of LP Problems

LP problems can always be converted to an equivalent standard form.

For example,

1

why it's not standard form.

$$
\begin{aligned}
\text{minimize} \quad & -2x_1 + 3x_2 \\
\text{subject to} \quad & x_1 + x_2 = 7, \\
& x_1 - 2x_2 \leq 4, \\
& x_1 \geq 0, \quad x_2 \in \mathbb{R}
\end{aligned}
$$

2

3

For example,

$$\begin{aligned} \text{minimize} \quad & -2x_1 + 3x_2 \\ \text{subject to} \quad & x_1 + x_2 = 7, \\ & x_1 - 2x_2 \leq 4, \\ & x_1 \geq 0, \end{aligned}$$

negate.

is equivalent to

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2 \\ \text{subject to} \quad & x_1 + x_2 = 7, \\ & x_1 - 2x_2 \leq 4, \\ & x_1 \geq 0, \end{aligned}$$

$x_2 \in \mathbb{R}$

Replace by $x_2' - x_2''$ s.t.

$x_2', x_2'' \geq 0.$

$x_1 + x_2 \leq 7$

$x_1 + x_2 \geq 7.$

# Standardization of LP Problems

For example,

$x_2 = x_2' - x_2''$

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2 \\ \text{subject to} \quad & x_1 + x_2 = 7, \\ & x_1 - 2x_2 \leq 4, \\ & x_1 \geq 0, \end{aligned}$$

is equivalent to

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\ \text{subject to} \quad & x_1 + x_2' - x_2'' = 7 \\ & x_1 - 2x_2' + 2x_2'' \leq 4 \\ & x_1, x_2', x_2'' \geq 0. \end{aligned}$$

# Standardization of LP Problems

For example,

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\ \text{subject to} \quad & x_1 + x_2' - x_2'' = 7 \\ & x_1 - 2x_2' + 2x_2'' \leq 4 \\ & x_1, x_2', x_2'' \geq 0. \end{aligned}$$

is equivalent to

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\ \text{subject to} \quad & x_1 + x_2' - x_2'' \leq 7 \\ & x_1 + x_2' - x_2'' \geq 7 \\ & x_1 - 2x_2' + 2x_2'' \leq 4 \\ & x_1, x_2', x_2'' \geq 0. \end{aligned}$$

# Standardization of LP Problems

For example,

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\ \text{subject to} \quad & x_1 + x_2' - x_2'' \leq 7 \\ & x_1 + x_2' - x_2'' \geq 7 \\ & x_1 - 2x_2' + 2x_2'' \leq 4 \\ & x_1, x_2', x_2'' \geq 0. \end{aligned}$$

is equivalent to

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\ \text{subject to} \quad & x_1 + x_2' - x_2'' \leq 7 \\ & -x_1 - x_2' + x_2'' \leq -7 \\ & x_1 - 2x_2' + 2x_2'' \leq 4 \\ & x_1, x_2', x_2'' \geq 0. \end{aligned}$$

# Standardization of LP Problems

For example,

$x_2$ $x_3$

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2' + 3x_2'' \\
\text{subject to} \quad & x_1 + x_2' - x_2'' \leq 7 \\
& -x_1 - x_2' + x_2'' \leq -7 \\
& x_1 - 2x_2' + 2x_2'' \leq 4 \\
& x_1, x_2', x_2'' \geq 0.
\end{aligned}
$$

is equivalent to

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\
\text{subject to} \quad & x_1 + x_2 - x_3 \leq 7 \\
& -x_1 - x_2 + x_3 \leq -7 \\
& x_1 - 2x_2 + 2x_3 \leq 4 \\
& x_1, x_2, x_3 \geq 0.
\end{aligned}
$$

After converting the following to the standard form

$$\begin{aligned} \text{minimize} \quad & -2x_1 - 3x_2 \\ \text{subject to} \quad & -x_1 + 5x_2 = -5 \\ & x_1 \geq 0 \end{aligned}$$

$$\boxed{\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ & -x_1 + 5x_2 = -5 \\ & x_1 \geq 0 \end{aligned}}$$

how many variables will it have?

$$x_2 = \underline{x_2' - x_2''}$$

$$\begin{aligned} \max \quad & 2x_1 + 3(x_2' - x_2'') \\ & -x_1 + 5(x_2' - x_2'') = -5 \\ & x_1, x_2', x_2'' \geq 0 \end{aligned} \iff \begin{cases} -x_1 + 5x_2' - 5x_2'' \leq -5 \\ x_1 - 5x_2' + 5x_2'' \leq 5 \end{cases}$$

# Back to Equality Constraints

To apply the *simplex method*, we need the nonnegativity constraints to be the only constraints.

# Back to Equality Constraints

To apply the *simplex method*, we need the nonnegativity constraints to be the only constraints.

Thus, for each constraint in the form of

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i,$$

we add a slack variable $x_{n+i} \geq 0$ to convert the constraints to

$$x_{n+i} = b_i - \sum_{j=1}^{n} a_{ij} x_j,$$

$$x_{n+i} \geq 0$$

# Example of Slack Variables

Standard Form —

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\ \text{subject to} \quad & x_1 + x_2 - x_3 \leq 7 \\ & -x_1 - x_2 + x_3 \leq -7 \\ & x_1 - 2x_2 + 2x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Slack Form —

$$\begin{aligned} \text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\ \text{subject to} \quad & x_4 = 7 - x_1 - x_2 + x_3 \\ & x_5 = -7 + x_1 + x_2 - x_3 \\ & x_6 = 4 - x_1 + 2x_2 - 2x_3 \\ & x_1, x_2, \ldots, x_6 \geq 0. \end{aligned}$$

In the slack form —

- The variables on the left hand side (LHS) are called basic variables.
- Those on the right hand side (RHS) are called nonbasic variables.

# Slack Form

To be more concise, we often write

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\
\text{subject to} \quad & x_4 = 7 - x_1 - x_2 + x_3 \\
& x_5 = -7 + x_1 + x_2 - x_3 \\
& x_6 = 4 - x_1 + 2x_2 - 2x_3 \\
& x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.
\end{aligned}
$$

LP → Standard form

↓

Slack form.

simply as

$$
\boxed{
\begin{aligned}
z &= 2x_1 - 3x_2 + 3x_3 \\
x_4 &= 7 - x_1 - x_2 + x_3 \\
x_5 &= -7 + x_1 + x_2 - x_3 \\
x_6 &= 4 - x_1 + 2x_2 - 2x_3.
\end{aligned}
}
$$

This is called the slack form.

# Write Slack Form with a 6-Tuple

A LP problem in slack form

$$z = v + \sum_{j \in N} c_j x_j$$
$$x_i = b_i - \sum_{j \in N} a_{ij} x_j, \quad \text{for} \quad i \in B$$

can also be written as the 6-tuple $(N, B, A, b, c, v)$ where

- $N$ is the set of indices of nonbasic variables

🍰 What is $(N, B, A, b, c, v)$ for the following problem?

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$
$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$
$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$
$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2},$$

$N = \{3, 5, 6\}$

# Write Slack Form with a 6-Tuple

A LP problem in slack form

$$z = v + \sum_{j \in N} c_j x_j$$
$$x_i = b_i - \sum_{j \in N} a_{ij} x_j, \quad \text{for} \quad i \in B$$

can also be written as the 6-tuple $(N, B, A, b, c, v)$ where

- $N$ is the set of indices of nonbasic variables
- $B$ is the set of indices of basic variables

What is $(N, B, A, b, c, v)$ for the following problem?

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$
$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$
$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$
$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2},$$

$B = \{1, 2, 4\}$.

# Write Slack Form with a 6-Tuple

A LP problem in slack form

$$z = v + \sum_{j \in N} c_j x_j$$

$$x_i = b_i - \sum_{j \in N} a_{ij} x_j, \quad \text{for} \quad i \in B$$

can also be written as the 6-tuple $(N, B, A, b, c, v)$ where

- $N$ is the set of indices of nonbasic variables
- $B$ is the set of indices of basic variables
- $A = (a_{ij})$ is the matrix of coefficients

🍰 What is $(N, B, A, b, c, v)$ for the following problem?

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$

$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$

$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$

$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2},$$

$b$

$$A = \begin{bmatrix} -\frac{1}{6} & -\frac{1}{6} & \frac{1}{3} \\ \frac{8}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix}$$

# Write Slack Form with a 6-Tuple

A LP problem in slack form

$$z = v + \sum_{j \in N} c_j x_j$$
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- $N$ is the set of indices of nonbasic variables
- $B$ is the set of indices of basic variables
- $A = (a_{ij})$ is the matrix of coefficients
- $b = (b_i)$ is a vector

🍰 What is $(N, B, A, b, c, v)$ for the following problem?

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$
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# Write Slack Form with a 6-Tuple

A LP problem in slack form

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- $N$ is the set of indices of nonbasic variables
- $B$ is the set of indices of basic variables
- $A = (a_{ij})$ is the matrix of coefficients
- $b = (b_i)$ is a vector
- $v$ is a constant

🍰 What is $(N, B, A, b, c, v)$ for the following problem?

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$
$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$
$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$
$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2},$$

# What are your main takeaways today? Any questions?